A. V. PAREKH TECHNICAL INSTITUTE, RAJKOT
COMPUTER ENGINEERING DEPARTMENT

Unit – IV

Cloud Computing
(Course Code: 4360709)

Virtualization

Vision Mission

Vision of Computer Department

Develop globally competent Computer Engineering Professionals who achieve excellence in an environment conducive for technical knowledge, skills, moral and ethical values with a focus to serve the society.

Mission of Computer Department

M1 : To provide state of the art infrastructure and facilities for imparting quality education and computer engineering skills for societal benefit.

M2 : Adopt industry oriented curriculum with an exposure to technologies for building systems & application in computer engineering.

M3 : To provide quality technical professional as per the industry and societal needs, encourage entrepreneurship, nurture innovation and life skills in consonance with latest interdisciplinary trends.

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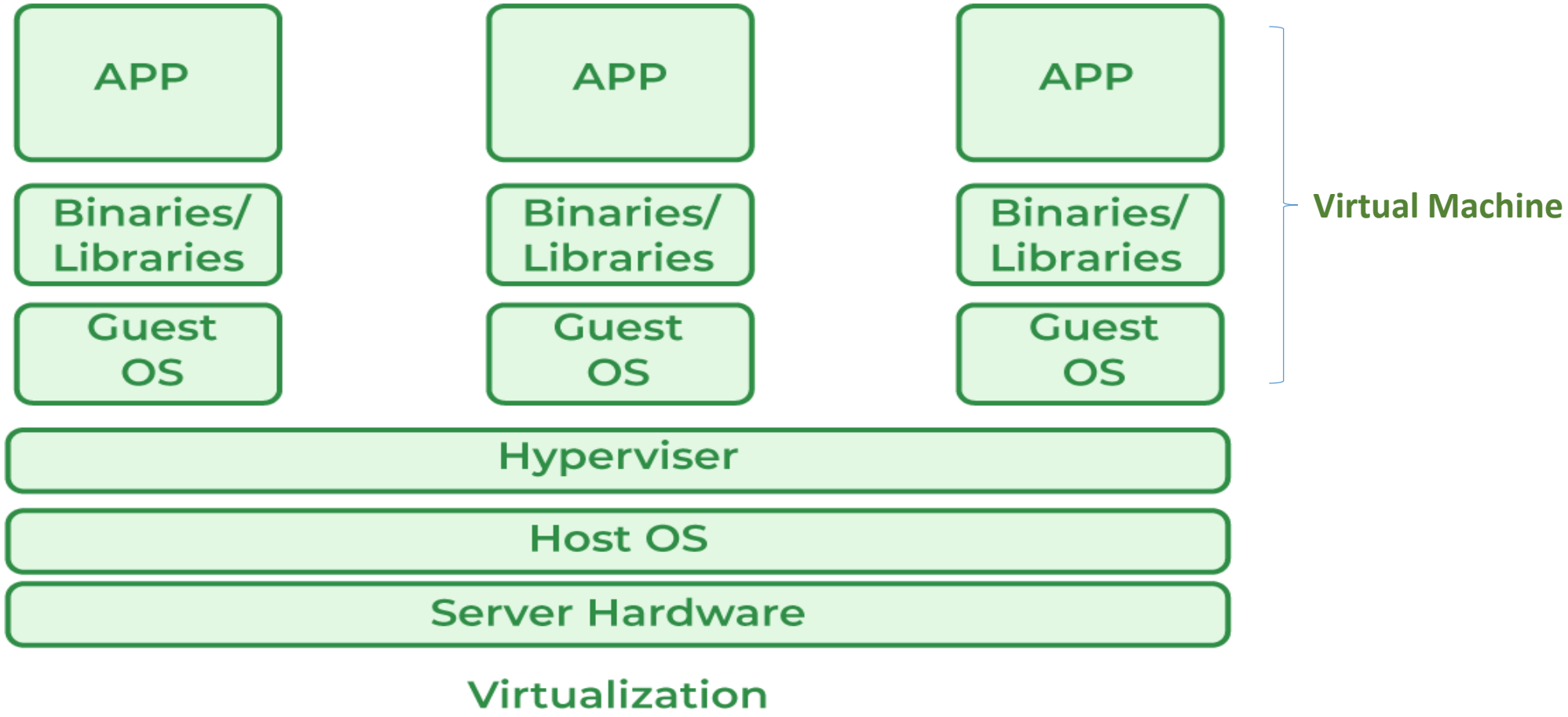
4.2.1 Technologies of virtualization : Hypervisors

Introduction of Virtualization

- **Virtualization** is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources".
- Formally, Virtualization is a technique, which allows sharing of a single physical instance of a resource or an application among multiple customers and organizations.
- This is done by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.
- virtualization technologies provide a virtual environment for not only executing applications but also for storage, memory, and networking.

- A Virtual machine provides an environment that is logically separated from the underlying hardware.
- The machine on which the virtual machine is going to create is known as **Host Machine** and that virtual machine is referred as a **Guest Machine.**



Levels of Implementing Virtualization

1. Instruction Set Architecture Level (ISA)
2. Hardware Abstraction Level (HAL)
3. Operating System Level
4. Library Level
5. Application Level

Five Levels of Virtualization

Application Level

JVM / .NET CLR

Library Level

WINE / vCUDA

Operating System Level

Virtual Environment / FVM

Hardware Abstraction Level

VMWare / Virtual PC

Instruction Set Architecture Level

BIRD / Dynamo

Instruction Set Architecture Level (ISA)

In ISA, virtualization works through an ISA emulation. This is helpful to run heaps of legacy code which was originally written for different hardware configurations.

- These codes can be run on the virtual machine through an ISA.
- A binary code that might need additional layers to run can now run on an x86 machine or with some tweaking, even on x64 machines. ISA helps make this a hardware-agnostic virtual machine.
- The basic emulation, though, requires an interpreter. This interpreter interprets the source code and converts it to a hardware readable format for processing.

Hardware Abstraction Level (HAL)

- As the name suggests, this level helps perform virtualization at the hardware level. It uses a bare hypervisor for its functioning.
- This level helps form the virtual machine and manages the hardware through virtualization.
- It enables virtualization of each hardware component such as I/O devices, processors, memory, etc.
- This way multiple users can use the same hardware with numerous instances of virtualization at the same time.

Operating System Level

- At the operating system level, the virtualization model creates an abstract layer between the applications and the OS.
- It is like an isolated container on the physical server and operating system that utilizes hardware and software. Each of these containers functions like servers.
- When the number of users is high, and no one is willing to share hardware, this level of virtualization comes in handy.
- Here, every user gets their own virtual environment with dedicated virtual hardware resources. This way, no conflicts arise.

Library Level

- OS system calls are lengthy and cumbersome. Which is why applications opt for APIs from user-level libraries.
- Most of the APIs provided by systems are rather well documented. Hence, library level virtualization is preferred in such scenarios.
- Library interfacing virtualization is made possible by API hooks. These API hooks control the communication link from the system to the applications.
- Some tools available today, such as vCUDA and WINE, have successfully demonstrated this technique.

Application Level

- Application-level virtualization comes handy when you wish to virtualize only an application. It does not virtualize an entire platform or environment.
- On an operating system, applications work as one process. Hence it is also known as process-level virtualization.
- It is generally useful when running virtual machines with high-level languages. Here, the application sits on top of the virtualization layer, which is above the application program.
- The application program is, in turn, residing in the operating system.
- Programs written in high-level languages and compiled for an application-level virtual machine can run fluently here.

Types of Virtualization

1. Desktop virtualization
2. Network Virtualization
3. Storage Virtualization
4. Data virtualization
5. Hardware Virtualization
6. Software Virtualization

Desktop virtualization

- Desktop virtualization allows the users' OS to be remotely stored on a server in the data center.
- It allows the user to access their desktop virtually, from any location by a different machine.
- desktop virtualization allows creating and storing different user desktop instances on a single host
- It is achieved by using a hypervisor (virtual machine monitor), which lives on top of the host server hardware to run and allows virtual desktops to use the computing power of the basic server hardware.
- Users who want specific operating systems other than Windows Server will need to have a virtual desktop.
- The main benefits of desktop virtualization are user mobility, portability, and easy management of software installation, updates, and patches.

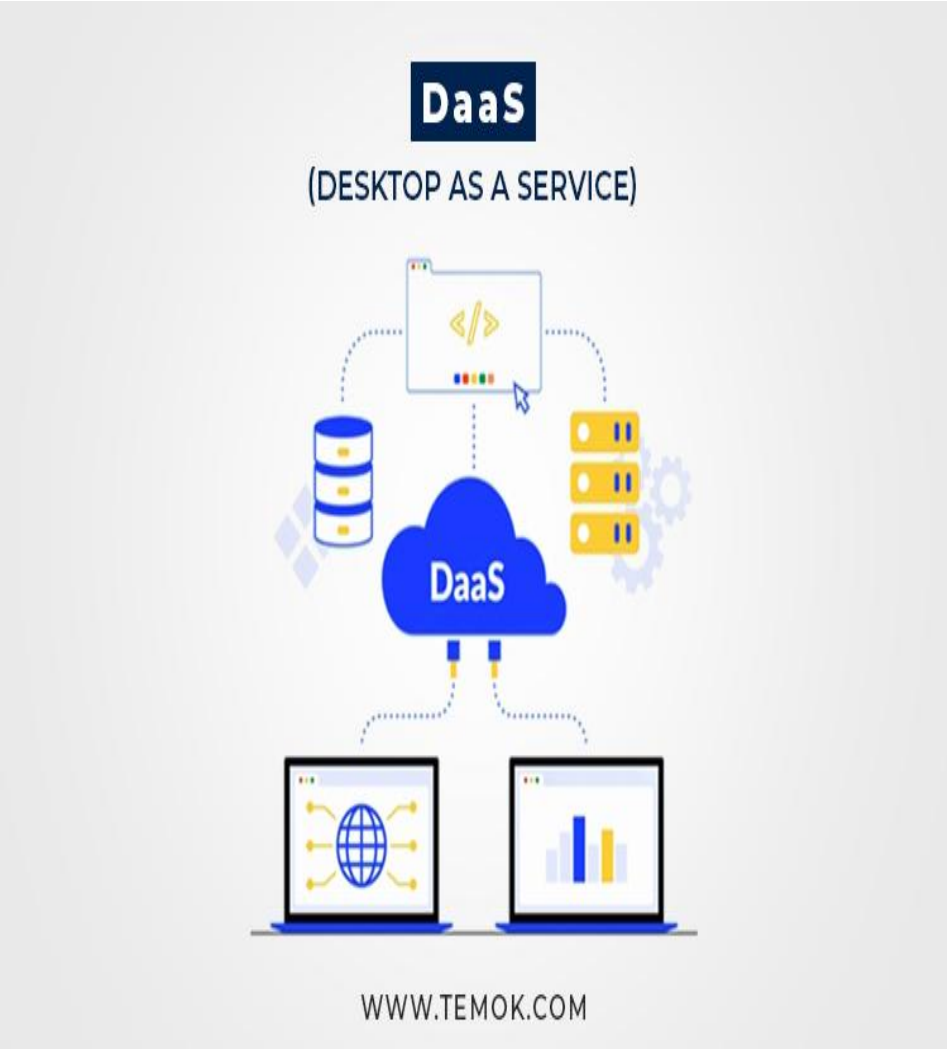
Desktop virtualization

- **Benefits:**

1. Simpler Administration
2. Saves Money
3. Increases Productivity
4. Security
5. Agility And Scalability
6. Better End-User Experiences
7. Supported Devices

- **Example:**

Azure virtual Desktop



Network Virtualization:

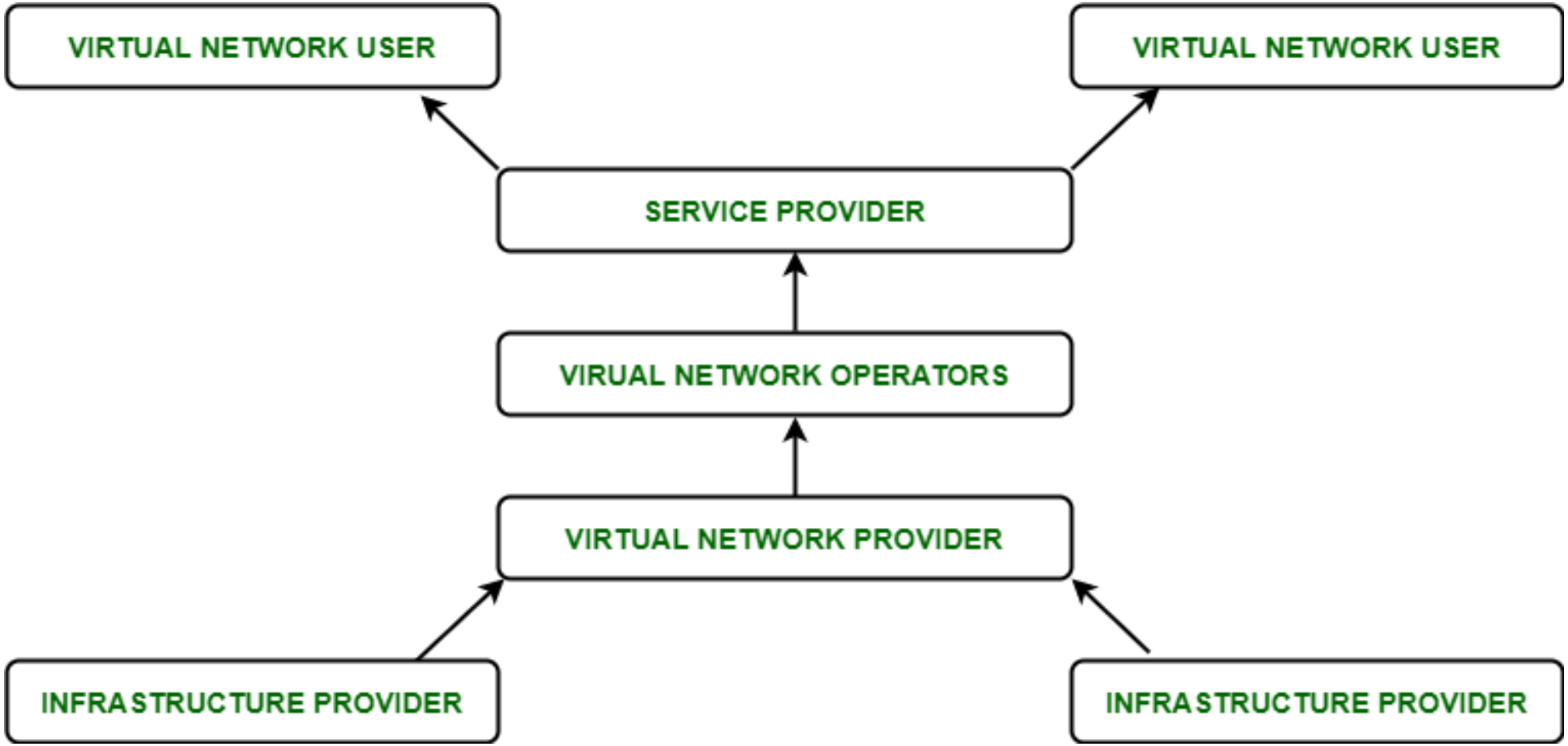
- The ability to run multiple virtual networks with each having a separate control and data plan.
- It co-exists together on top of one physical network. It can be managed by individual parties that are potentially confidential to each other.
- Network virtualization provides a facility to create and provision virtual networks, logical switches, routers, firewalls, load balancers, Virtual Private Networks (VPN), and workload security within days or even weeks.

Network Virtualization:

- **Functions of Network Virtualization :**

1. It enables the functional grouping of nodes in a virtual network.
2. It enables the virtual network to share network resources.
3. It allows communication between nodes in a virtual network without routing of frames.
4. It restricts management traffic.
5. It enforces routing for communication between virtual networks.

General Architecture Of Network Virtualization:



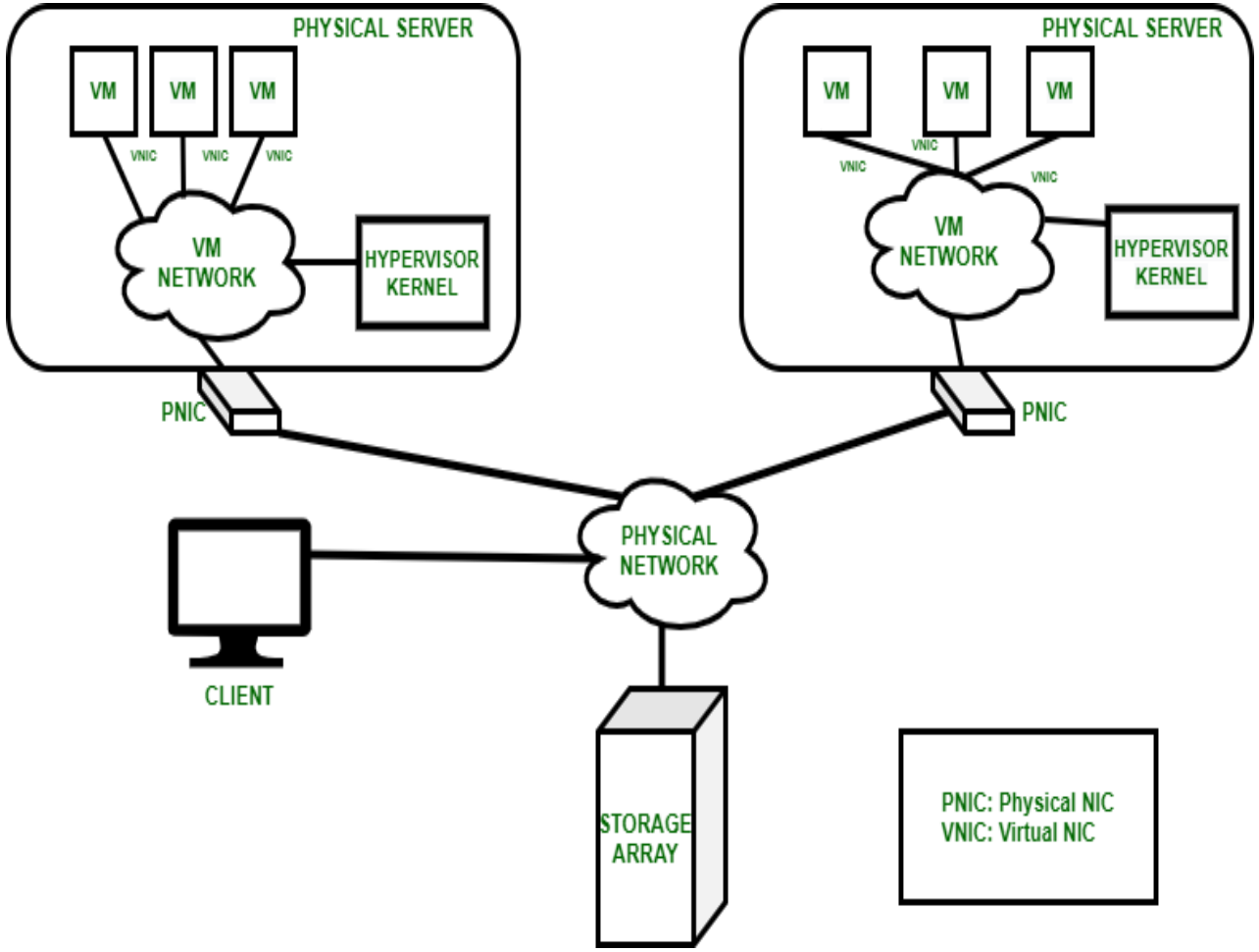
Network Virtualization in data center:

- **Physical Network**

1. Physical components: Network adapters, switches, bridges, repeaters, routers and hubs.

- **VM Network**

1. Consists of virtual switches.
2. Provides connectivity to hypervisor kernel.
3. Connects to the physical network.
4. Resides inside the physical server.



Advantages of Network Virtualization :

- **Improves manageability –**
  - Grouping and regrouping of nodes are eased.
  - Configuration of VM is allowed from a centralized management workstation using management software.
- **Reduces CAPEX –(Capital Expenditure)**
  - The requirement to set up separate physical networks for different node groups is reduced.
- **Improves utilization –**
  - Multiple VMs are enabled to share the same physical network which enhances the utilization of network resource.
- **Enhances performance –**
  - Network broadcast is restricted and VM performance is improved.
- **Enhances security –**
  - Sensitive data is isolated from one VM to another VM.
  - Access to nodes is restricted in a VM from another VM.

Disadvantages of Network Virtualization :

- It needs to manage IT in the abstract.
- It needs to coexist with physical devices in a cloud-integrated hybrid environment.
- Increased complexity.
- Upfront cost.
- Possible learning curve.

- **Examples of Network Virtualization :**
- VLAN
- **Network Overlays –**
  - provided by an encapsulation protocol called VXLAN for overlaying virtualized layer 2 networks over layer 3 networks
- **Network Virtualization Platform: VMware NSX –**

Application of Network Virtualization :

- Network virtualization may be used in the development of application testing to mimic real-world hardware and system software.
- It helps us to integrate several physical networks into a single network or separate single physical networks into multiple analytical networks.
- In the field of application performance engineering, network virtualization allows the simulation of connections between applications, services, dependencies, and end-users for software testing.
- It helps us to deploy applications in a quicker time frame, thereby supporting a faster go-to-market.
- Network virtualization helps the software testing teams to derive actual results with expected instances and congestion issues in a networked environment.

Storage Virtualization

- Storage virtualization is an array of servers that are managed by a virtual storage system.
- The servers aren’t aware of exactly where their data is stored and instead function more like worker bees in a hive.
- It makes managing storage from multiple sources be managed and utilized as a single repository.
- Storage virtualization software maintains smooth operations, consistent performance, and a continuous suite of advanced functions despite changes, breaks down, and differences in the underlying equipment.

Storage Virtualization

- Storage virtualization is becoming more and more important in various other forms:

1. **File servers:** The operating system writes the data to a remote location with no need to understand how to write to the physical media.
2. **WAN Accelerators:** Instead of sending multiple copies of the same data over the WAN environment, WAN accelerators will cache the data locally and present the re-requested blocks at LAN speed, while not impacting the WAN performance.
3. **SAN and NAS:** Storage is presented over the Ethernet network of the operating system. NAS(network attached storage) presents the storage as file operations . SAN(Storage Area Network) technologies present the storage as block level storage (like Fibre Channel). SAN technologies receive the operating instructions only when if the storage was a locally attached device.
4. **Storage Tiering:** Utilizing the storage pool concept as a stepping stone, storage tiering analyze the most commonly used data and places it on the highest performing storage pool. The lowest one used data is placed on the weakest performing storage pool.

Advantages Storage Virtualization

1. Data is stored in the more convenient locations away from the specific host. In the case of a host failure, the data is not compromised necessarily.
2. The storage devices can perform advanced functions like replication, reduplication, and disaster recovery functionality.
3. By doing abstraction of the storage level, IT operations become more flexible in how storage is provided, partitioned, and protected.
4. Usability or Accessibility
5. A cloud storage service may be used by multiple users, and as everything is handled and automated by the cloud provider vendor, one user’s current task would not influence that of another.
6. You can upgrade the service plan if the storage included in the current plan is insufficient.

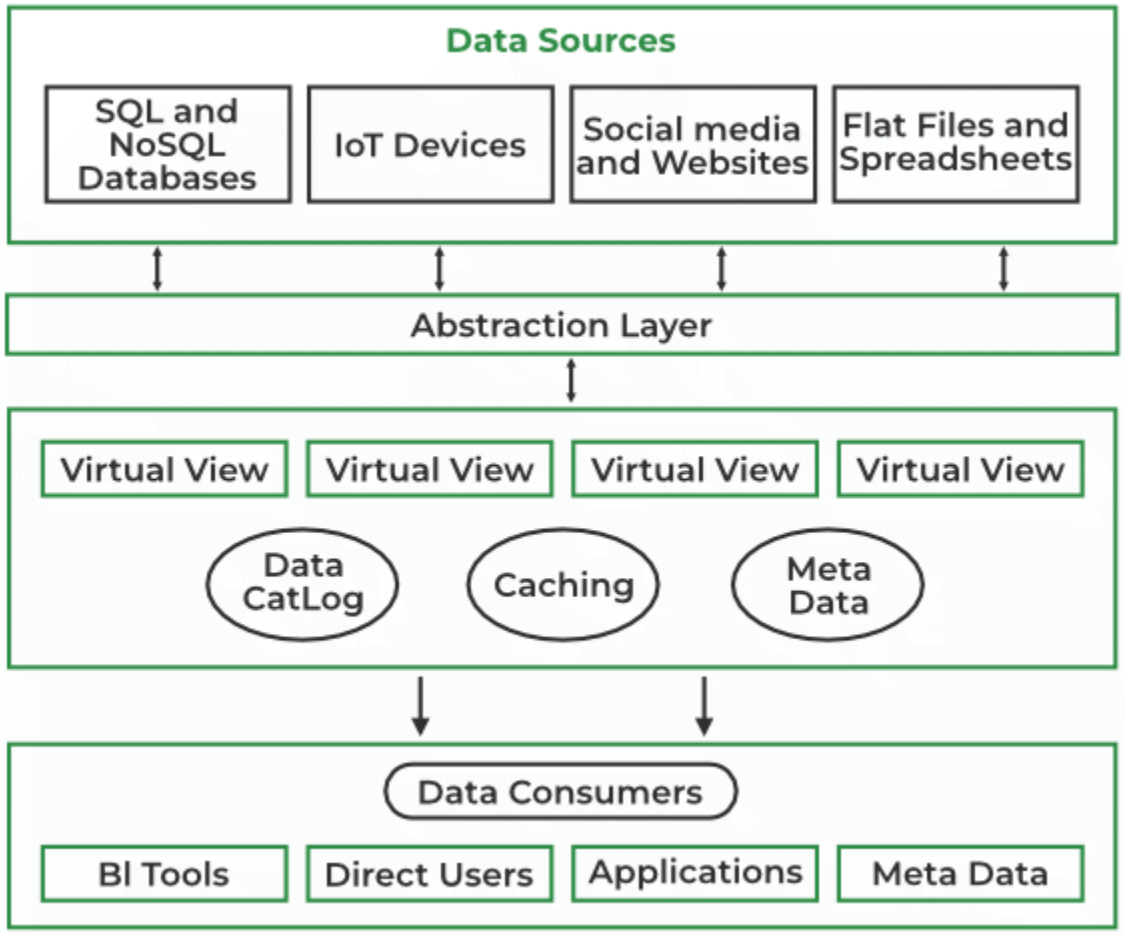
Disadvantages Storage Virtualization

1. **Vulnerability:** The majority of PCs and servers that save data require an internet connection. hus, making them vulnerable to attacks by malicious users on the network.
2. **Internet Dependency:**
3. **Issues in Security and Privacy:** Confidential data must be given over to a third-party organization in order to be stored in the cloud. One must therefore have complete faith in the cloud vendor.
4. **Limitations on Control:** After the user moves data to the cloud, the vendor is now in charge of it. This implies that users must rely on the vendors to maintain their services in a safe, stable, up-and-running, and fully functional manner.
5. **Cost:** For small short-term and small scale projects term project it is not feasible
6. **Migration:** one may choose to migrate data to another solution if one discovers that the cloud provider does not adequately address all of the users' needs
7. **Minimal Support:** Even when using a simple, rapid, and secure cloud solution, problems can occur because every user and business has different wants and specifics.
8. **Data Management:** Given that cloud storage systems have their organizational systems, managing data may be a pain. The system used by a cloud vendor might not work with the way one manages their storage currently.

Data virtualization

- This is the kind of virtualization in which the data is collected from various sources and managed at a single place.
- It is not known more about the technical information like how data is collected, stored & formatted then arranged that data logically so that its virtual view can be accessed by its interested people and stakeholders, and users through the various cloud services remotely.
- Many big giant companies are providing their services like Oracle, IBM, At scale, Cdata, etc.

Data virtualization



Layers of Data virtualization

- **Connection layer:** With the use of connectors and communication protocols, this layer is in charge of accessing the data dispersed across numerous source systems that contain both organized and unstructured data. Platforms for data virtualization can connect to various data sources, such as SQL and NoSQL databases like MySQL, Oracle and MongoDB etc.
- **Abstraction layer:** The abstraction layer, also known as the virtual or semantic layer, serves as a link between all data sources and all business users, forming the backbone of the entire virtualization system. This tier just holds the logical views and information required to access the sources; it does not itself store any data. The complexity of the underlying data structures is hidden from end users, who only see the schematic data models thanks to the abstraction layer.
- **Consumption layer:** A single point of access to the data stored in the underlying sources is offered by a different tier of the data virtualization architecture. Depending on the type of consumer, several protocols and connectors are used to give abstracted data representations. They can interface with the virtual layer using SQL and a variety of APIs, such as REST and SOAP APIs, as well as access standards like JDBC and ODBC. A variety of corporate users, tools, and apps can all have access to data virtualization software, including well-known ones like Tableau, Cognos, and Power BI.

Features of Data virtualization

- Time to market acceleration from data to final product
  - Virtual data objects can be created considerably more quickly than existing ETL(Extract-Transform-Load) tools and databases since they include integrated data.
- One-Stop Security
  - Authorizing numerous user groups on the same virtual dataset is feasible by using data masking, anonymization, and pseudonymization.
- Combine data explicitly from different sources
  - The virtual data layer makes it simple to incorporate distributed data from Data Warehouses, Big Data Platforms, Data lakes, Cloud Solutions, and Machine Learning into user-required data objects.
- Flexibility
  - By providing integrated virtual data objects, data virtualization enables you to reply instantly to fresh data requests. This does away with the necessity to copy data to various data levels but just makes it virtually accessible.

Applications of Data Virtualization

- **Migration:** a gradual migration of old systems to the cloud. You can accomplish this with data virtualization without halting operations or reporting.
- **Uses In Operations:-** Data virtualization that spans data silos enables everyone, from a call center to a database manager, to see the full range of data repositories from a single point of access.
- **Agile BI:-** With data virtualization, you can use your data for data science, API or system linkages, governed (regulated), and self-service BI
- **Data Integration:-** This is the most likely situation you will encounter because practically every company contains data from multiple different data sources.
- **Accessing Real-Time Data:-** Data virtualization allows you to blend real-time data from the source system with historical data that has been "offloaded" to a different source. (This way one can save storage and caching)

Advantages of Data Virtualization

- Data virtualization enables real-time access to and manipulation of source data through the virtual/logical layer without physically relocating the data to a new location. ETL is typically not required.
- Comparing the implementation of data virtualization to the construction of a separate consolidated store, the former takes less funding and resources.
- There is no need to relocate the material, and access levels may be controlled.
- Without worrying about a data type or where the data is located, users can build and execute whatever reports and analyses they require.
- Through a single virtual layer, all corporate data is accessible to all consumers and use cases.

Examples of Data Virtualization

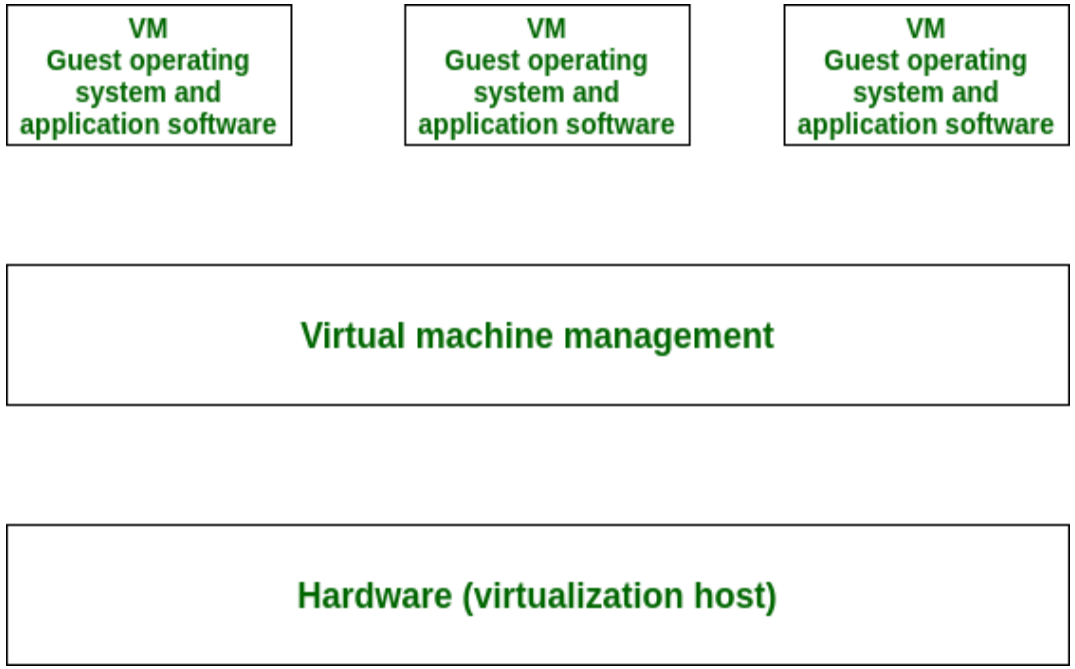
- Red Hat JBoss data virtualization
- TIBCO data virtualization
- Oracle data service integrator
- SAS Federation Server
- Denodo

Industries that use Data Virtualization

- Communication & Technology
- Finance
- Government
- Healthcare
- Manufacturing

Hardware Virtualization

- When the virtual machine software or virtual machine manager *(VMM) is directly installed on the hardware system* is known as hardware virtualization.
- The main job of hypervisor is to control and monitoring the processor, memory and other hardware resources.
- After virtualization of hardware system we can install different operating system on it and run different applications on those OS.
- Its mostly implemented for the server platforms, because controlling virtual machines is much easier than controlling a physical server.



features of hardware-based virtualization

- **Isolation:** Hardware-based virtualization provides strong isolation between virtual machines, which means that any problems in one virtual machine will not affect other virtual machines running on the same physical host.
- **Security:** Hardware-based virtualization provides a high level of security as each virtual machine is isolated from the host operating system and other virtual machines, making it difficult for malicious code to spread from one virtual machine to another.
- **Performance:** Hardware-based virtualization provides good performance as the hypervisor has direct access to the physical hardware, which means that virtual machines can achieve close to native performance.
- **Resource allocation:** Hardware-based virtualization allows for flexible allocation of hardware resources such as CPU, memory, and I/O bandwidth to virtual machines.

features of hardware-based virtualization

- **Snapshot and migration:** Hardware-based virtualization allows for the creation of snapshots, which can be used for backup and recovery purposes. It also allows for live migration of virtual machines between physical hosts, which can be used for load balancing and other purposes.
- **Support for multiple operating systems:** Hardware-based virtualization supports multiple operating systems, which allows for the consolidation of workloads onto fewer physical machines, reducing hardware and maintenance costs.
- **Compatibility:** Hardware-based virtualization is compatible with most modern operating systems, making it easy to integrate into existing IT infrastructure.

Advantages of hardware-based virtualization

- It reduces the maintenance overhead of paravirtualization as it reduces (ideally, eliminates) the modification in the guest operating system.
- It is also significantly convenient to attain enhanced performance.

Disadvantages of hardware-based virtualization

- Hardware-based virtualization requires explicit support in the host CPU, which may not be available on all x86/x86_64 processors.
- A “pure” hardware-based virtualization approach, including the entire unmodified guest operating system, involves many VM traps, and thus a rapid increase in CPU overhead occurs which limits the scalability and efficiency of server consolidation.

Examples of Tools

- Hyper V
- Vmware vsphere
- Oracle VM Virtual Box
- Citrix Hypervisor

Software Virtualization

- *Software Virtualization is able to abstract the software installation procedure and create virtual software installations*.
- **Virtualized software** is an application that will be "installed" into its own self-contained unit.
- Example of software virtualization is *VMware software, virtual box* etc.

Benefits of Virtualization

- More flexible and efficient allocation of resources.
- Enhance development productivity.
- It lowers the cost of IT infrastructure.
- Remote access and rapid scalability.
- High availability and disaster recovery.
- Pay peruse of the IT infrastructure on demand.
- Enables running multiple operating systems.

Drawback of Virtualization

- **High Initial Investment:** Clouds have a very high initial investment, but it is also true that it will help in reducing the cost of companies.
- **Learning New Infrastructure:** As the companies shifted from Servers to Cloud, it requires highly skilled staff who have skills to work with the cloud easily, and for this, you have to hire new staff or provide training to current staff.
- **Risk of Data:** Hosting data on third-party resources can lead to putting the data at risk, it has the chance of getting attacked by any hacker or cracker very easily.

Characteristics of Virtualization

- **Increased Security:** The ability to control the execution of a guest program in a completely transparent manner opens new possibilities for delivering a secure, controlled execution environment. All the operations of the guest programs are generally performed against the virtual machine, which then translates and applies them to the host programs.
- **Managed Execution:** In particular, sharing, aggregation, emulation, and isolation are the most relevant features.
- **Sharing:** Virtualization allows the creation of a separate computing environment within the same host.
- **Aggregation:** It is possible to share physical resources among several guests, but virtualization also allows aggregation, which is the opposite process.

Applications of Virtualization

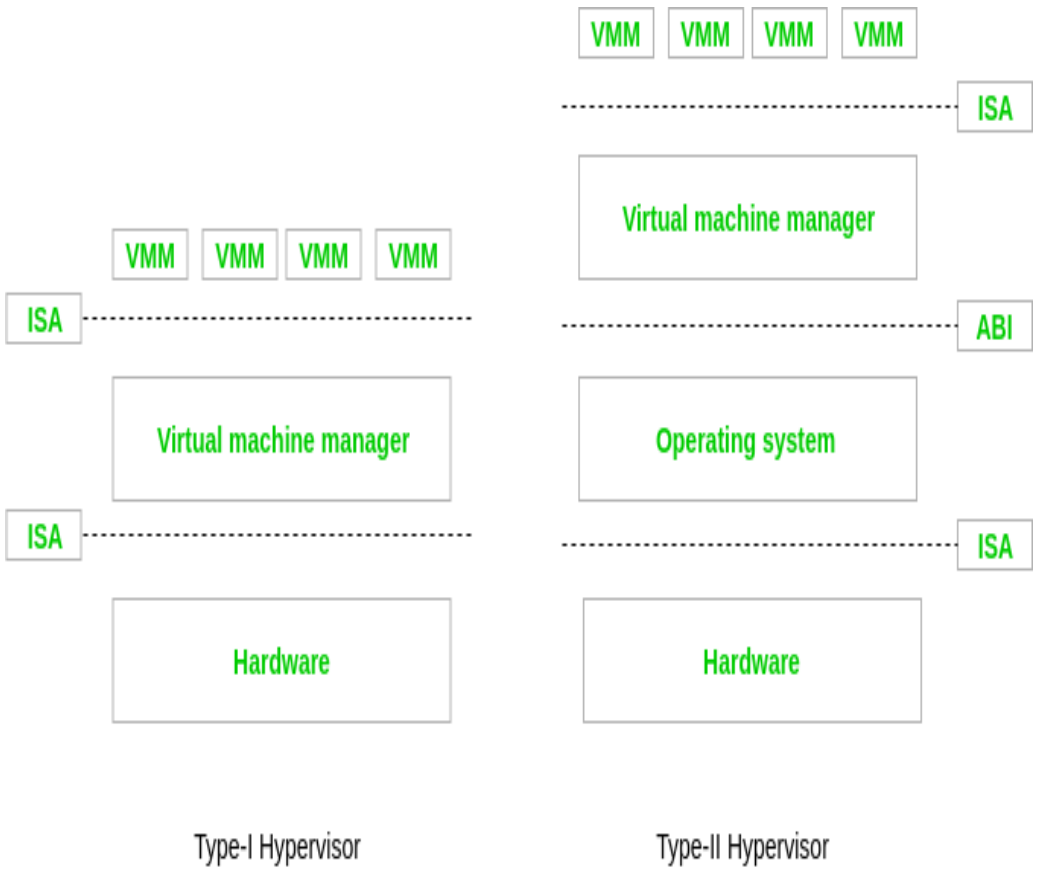
- Data-integration
- Business-integration
- Service-oriented architecture data-services
- Searching organizational data

Technologies of virtualization : Hypervisors

- A hypervisor, also known as a virtual machine monitor or VMM.
- The hypervisor is a piece of software that allows us to build and run virtual machines.
- The guest VMs are independent of the host hardware which is one of the major benefits of the Hypervisor.
- Thus they can be quickly switched between servers.
- It allows several virtual machines to operate on a single physical server.

Types of hypervisors

- There are two types of hypervisors: "Type 1" (also known as "bare metal") and "Type 2" (also known as "hosted").
- A type 1 hypervisor functions as a light operating system that operates directly on the host's hardware, while a type 2 hypervisor functions as a software layer on top of an operating system, similar to other computer programs.



Type 1 hypervisor

- Type 1 hypervisor replaces the host operating system, and the hypervisor schedules VM services directly to the hardware.
- As a result, they stand in for operating systems and communicate directly with the ISA(Instruction set Arch) interface offered by the underlying hardware
- The type 1 hypervisor is very much commonly used in the enterprise data center or other server-based environments.
- It includes KVM, Microsoft Hyper-V, and VMware vSphere. If we are running the updated version of the hypervisor then we must have already got the KVM integrated into the Linux kernel in 2007.

Type 2 hypervisor

- It is also known as a hosted hypervisor, The type 2 hypervisor is a software layer or framework that runs on a traditional operating system.
- Type II hypervisors require the assistance of an operating system. This means they're operating system-managed applications that communicate with it via the ABI(Application Binary Interface) and simulate the ISA of virtual hardware for guest operating systems.
- It operates by separating the guest and host operating systems. The host operating system schedules VM services, which are then executed on the hardware.
- Individual users who wish to operate multiple operating systems on a personal computer should use a form 2 hypervisor.
- This type of hypervisor also includes the virtual machines with it.

Comparison on Type-1 and Type-2 Hypervisor

- Hardware acceleration technology improves the processing speed of both bare-metal and hosted hypervisors, allowing them to build and handle virtual resources more quickly.
- A type 1 hypervisor functions as a light operating system that operates directly on the host's hardware, while a type 2 hypervisor functions as a software layer on top of an operating system, similar to other computer programs.
- Bare-metal hypervisors are isolated from the attack-prone operating system thus they are extremely stable.
- Bare-metal hypervisors are usually faster and more powerful than hosted hypervisors. For these purposes, the majority of enterprise businesses opt for bare-metal hypervisors for their data center computing requirements.

- Hosted hypervisors run inside the OS, they can be topped with additional (and different) operating systems.
- The hosted hypervisors have longer latency than bare-metal hypervisors which is a very major disadvantage of the it. This is due to the contact between the hardware and the hypervisor must go through the OS's extra layer.

Choosing the right hypervisor (Xen, KVM)

- Xen is an open source hypervisor that runs directly on the system hardware.
- It's a type 1 hypervisor, also known as a bare-metal hypervisor that creates logical pools of system resources so that many virtual machines can share the same physical resources.
- Xen inserts a virtualization layer between the system hardware and the virtual machines.
- This allows Xen to dynamically allocate the system hardware to any guest operating system.

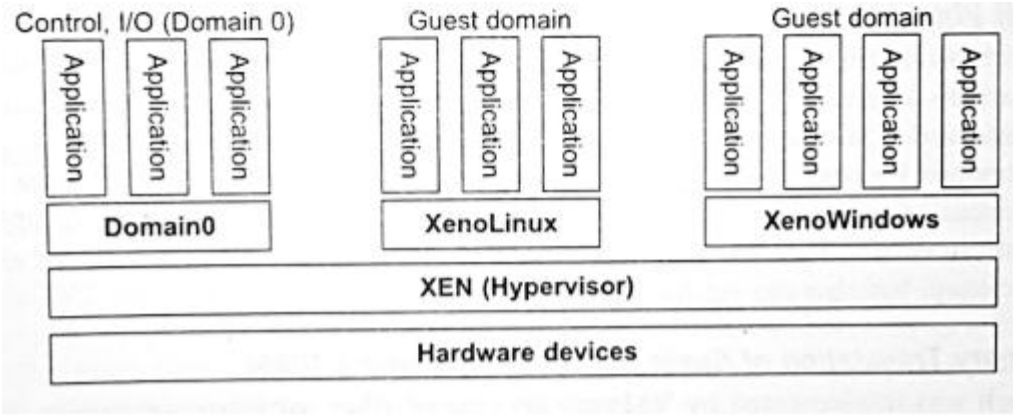


Fig. Xen Architecture

KVM (Kernel-based **Virtual** Machine)

- Kernel-based Virtual Machine (KVM) is an open source virtualization technology built into Linux®. Specifically, KVM lets you turn Linux into a hypervisor that allows a host machine to run multiple, isolated virtual environments called guests or virtual machines (VMs)
- KVM provides high performance and scalability due to its native support for hardware virtualization extensions and paravirtualization drivers. Furthermore, KVM has an abundant ecosystem of management tools, including libvirt, virt-manager, virsh, and more.
- KVM converts Linux into a type-1 (bare-metal) hypervisor. All hypervisors need some operating system-level components—such as a memory manager, process scheduler, input/output (I/O) stack, device drivers, security manager, a network stack, and more—to run VMs. KVM has all these components because it’s part of the Linux kernel. Every VM is implemented as a regular Linux process, scheduled by the standard Linux scheduler, with dedicated virtual hardware like a network card, graphics adapter, CPU(s), memory, and disks.

How do you choose between KVM and Xen?

- When selecting a hypervisor for Linux virtualization, there is no definitive answer as it depends on your use case, requirements, and preferences.
- However, KVM is a good choice if you want to leverage the Linux kernel features for virtualization and containers, run a wide range of guest operating systems, use various management tools and plugins, and achieve high performance and scalability.
- Alternatively, Xen is a good option if you’re looking for a bare-metal hypervisor with high isolation and security, reduced overhead and complexity, and a mature and stable code base.